The Open Doors of a Dark-Energy-Tied Galaxy Law: Scoped No-Gos, Record Corrections, and a Cassini Split between the Unscreened Law and the Screened Theory

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Abstract

The owner's programme studies a gravity framework whose galactic law is $g_N = g^2/(g+a_0)$ with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ and $\kappa = \frac{1}{2}$ fitted. Its extra gravitating mass is a collisionless "cold energy" medium, not a particle. On 2026-10-22 we made one rigorous pass over every door the record had left open (lanes D1–D9b), followed by an audit of hidden inputs (wave E1–E5). Each lane froze its criteria before computing and ran a mutation control. Both a_0 footings (9.36×10^{-11} and $1.13\times10^{-10}\,\mathrm{m\,s^{-2}}$) are reported and never pooled. Most results are scoped no-gos and corrections to the record. A finding that changes several others: the framework's own action gives AQUAL with $\mu = x/(1+x)$, not the QUMOND that three lanes used (E2).

1. **Cassini splits the theory in two.** The unscreened canonical law fails Cassini Q_2 by 2.44–4.59× the Hees et al. (2014) 2σ line (canonical) and 2.85–5.19× (alternative) with the action's own field equation (QUMOND: 1.70–3.70× and 1.91–4.03×). The strong-field turn-off cannot help: across its whole band it moves Q_2 by 0.052σ. This is the RAR–Q_2 class conflict of Desmond, Hees & Famaey (2024). The screened theory (stiffness length $L_\star = 24$–$90\,\mathrm{pc}$) passes: its Q_2 is at most 2.5×10^{-8} of the Hees line, because inside $L_\star$ the Sun's field is a hedgehog with no quadrupole; it passes for every $L_\star$ above a threshold that is at most 0.069 pc (canonical) or 0.111 pc (alternative), an upper bound on the threshold.
2. **The screened theory is the one version that passes both Cassini and the galactic law.** Its open items, stated plainly: the satellite dwarfs sit $+0.89\pm0.13$ dex above its prediction, and the cold mass needed is supplied only through a ΛCDM-processed census with no binding mechanism in the record (E3); $L_\star$ is soft (native floor 28.85 pc, 20.0–42.3 pc over the stellar M/L uncertainty) and u is an a_0- and h-dependent number (0.433 at the galaxy-measured a_0; E1, E4); no covariant parent exists for the stated classes, with or without stiffness (D9, D9b, E5). Gaia DR4 is the decisive test: the screened theory predicts $\gamma_v - 1 \approx 8.1\times10^{-6}$ for a 20 kAU binary, the unscreened law-first reading $\gamma_v \approx 1.17$.
3. **κ is not derived.** No horizon-thermodynamic monomial gives $\frac{1}{2}$ without square roots; with them, $\frac{1}{2}$ appears only as $\sqrt{G\rho_\Lambda}$ times a half picked from a menu. The class's natural output is Milgrom's $a_0 = cH_\Lambda/2\pi$, a postulate here. The data do not select $R = 4$ ($\kappa = \frac{1}{2}$) among natural caps: 5 of 7 grid values lie within 2σ. The measured κ is robust to the background (E4); the alternative footing is a pure ΛCDM import (E1).
4. **The unscreened law family.** No member passes both Q_2 and SPARC on both footings under AQUAL (E2). On the native background with QUMOND some members passed (E1); AQUAL raises their Q_2, so their status is knife-edge, not settled. The turn-off has no mechanism in the action, and its scale y_t is κ relabelled.
5. **No covariant parent.** A U(1)-dielectric parent that enforces the law's Gauss identity is superluminal or needs a dark-sector Lorentz violation of 6.3×10^7 at 1 AU; enforcing the identity in the record's healthy action brings back the no-self-gravity theorem; the stiffness changes none of this (E5).
6. **Tests, inputs and record.** The skew-sign test of the law's form is retired; the wide-binary fork is decided by systematics, not sample size; the sealed ALMA tests are expected to be unscorable; the native 1024^3 shear run died of memory with no verdict. No tested mechanism gives the retained fraction $u \approx 0.47$ (13.0σ from the record's cluster model; inflow stream counting is rejected by the profile, $\chi^2 = 39.0$ for 8 degrees of freedom). Four earlier verdicts flip under clean inputs (E1). Six errata were written into the record.

1 Setting and rules

- **The law.** $g_N = g^2/(g+a_0)$, the record's canonical ($n = 1$) form. A tethered cold medium, whose excess density is the law's phantom, supplies the extra mass (PAPER57). Footings: canonical $a_0 = 9.36\times10^{-11}$, alternative $1.13\times10^{-10}\,\mathrm{m\,s^{-2}}$.
- **κ.** $\kappa = \frac{1}{2}$ is fitted. The thermal value $\kappa = Z/4\pi = 0.460659$ ($a_0 = cH_\Lambda/2\pi$, $Z = 2\sqrt{8\pi/3} = 5.78881$) is a postulate. It is Milgrom's relation $2\pi a_0 \approx c^2(\Lambda/3)^{1/2}$ (arXiv:1605.07458, eq. 5), with the de Sitter Unruh mechanism of Milgrom (1999). The record's DSF-21 documents were corrected to say so (commit d3e4110528).
- **Field equation (E2).** Varying the record's action (PAPER49; DSF-21 A33 at $L_\star \to 0$) gives $\nabla\cdot[\mu(|\nabla\Phi|/a_0)\nabla\Phi] = 4\pi G\rho$ with $\mu = x/(1+x)$: AQUAL, not QUMOND. The action makes $\nabla\Phi$ curl-free but not $\mu\nabla\Phi$; the divergence-free part is 22% of the anomalous field at r_M. D2, D7b and D7c used QUMOND; where E2 recomputed them, the AQUAL values supersede the QUMOND ones and both are quoted.

Table 1: The lanes of the doors push and of wave E. "Checks" is the main run; the mutation run must fail at least one. Commits: criteria / result. [a]Under QUMOND; see E1 and E2 for its status under clean inputs and the action's own field equation.

lane	door	verdict	checks	commits
D1	skew-sign test of $n = 1$ vs $n = 2$	SCOPED NO-GO	10/10	1f956de1e8 / ad9bea8355
D2	wide-binary fork	CRISP FINDING	7/7	c8960e6424 / c27b941b99
D3	CFG960 1024^3 run	CRISP FINDING (run)	–	77d59a9a38 (read-only)
D4	sealed ALMA tests	CRISP FINDING	6/6	3537e6f817 / f81efdcf6f
D5	why κ	SCOPED NO-GO	8/8	91ec2c9d07 / 962007715e
D6	retained fraction u	CRISP FINDING	6/7	3082c43c79, A1 daf92d2d11 / b21dda1a2d
D7	turn-off y_t	SCOPED NO-GO	11/11	f2e0423a60 / a2a07b53d9
D7b	Cassini Q_2, unscreened law	CRISP FINDING	7/8	c6a0b6f3c7 / 9542a09f54
D7c	Q_2–RAR gate, unscreened family	SCOPED NO-GO[a]	5/6	e9e5349368 / 26b5db080f
D7d	Cassini Q_2, screened theory	CRISP FINDING	10/10	a13e418498, A1 e1ba64cf92 / b26f14bb82
D8	value of $L_\star$	CRISP FINDING	5/5, 2/2	3327019734 / 3d5eb3ead9, f1da138742
D8b	universality of σ_c	CRISP FINDING	5/5	97ec7892fa / 8a9f0f1234
D9	covariant parent	SCOPED NO-GO	9/12 + 3/3	bfb93eb61c / 2edcee066c
D9b	A33 + multiplier	SCOPED NO-GO	14/14	ce12fb3210 / 61883a565f
E1	input ledger and contamination	CRISP FINDING	6/6	581d52210b / 77a8b2d4a6
E2	field equation of the action	CRISP FINDING	10/10	021ee82abb / 474050bcd8
E3	dwarf satellite excess	CRISP FINDING	6/6	5686635887 / c5e4196bb9
E4	the fitted three (κ, $L_\star$, u)	CRISP FINDING	6/6	9cc8fcae21, A1 9b3dfb1000 / c5a9768e6a
E5	parent logic with stiffness	SCOPED NO-GO	14/15	039c76d2f9, A1 ab0f5d006c / 08e3c1bfbc

- **The screened theory.** The same action with stiffness length $L_\star$ (the record's $L_\star \geq 24\,\mathrm{pc}$): galaxies follow the law, systems small next to $L_\star$ (the Sun, outer globulars, wide binaries) are screened toward Newton.
- **Method.** Each lane (directory `doors_push_2026_10_22/`) committed `FROZEN_CRITERIA.md` alone before any script, then a deterministic script whose output and `*_results.json` carry every number quoted here, and a mutation run that must break at least one check. Amendments are append-only and marked pre- or post-result. Table 1 lists the lanes.

2 Why κ? A no-go for the horizon class (D5)

The law gives $\rho_\Lambda c^2/(a_0^2/G) = 1/\kappa^2 \equiv R$, so $\kappa = \frac{1}{2}$ is $R = 4$, $\Lambda = 32\pi a_0^2/c^4$ and $a_0 = cH_\Lambda/Z$.

- **The class.** a_0 is any monomial in the de Sitter horizon quantities with their textbook coefficients (radius, area, volume, Gibbons–Hawking temperature, Bekenstein–Hawking entropy, Misner–Sharp, Komar and vacuum energies, surface gravity) times powers of $c, G, \hbar, k_B$, with the result free of $\hbar$ and k_B. Exponents are integers ($D = 1$) or half-integers ($D = 2$).
- **Theorem (checked on all 3,900 enumerated monomials).** Every member is $a_0 = q\pi^m cH_\Lambda$ with q^2 rational, so $\kappa^2 = (8/3)q^2\pi^{2m+1}$. For $D = 1$ the power of π is odd, so $\frac{1}{2}$ is impossible, and $Z/4\pi$ is the class's simplest member ($a_0 = ck_BT_{\mathrm{GH}}/\hbar$). For $D = 2$ all 16 constructions that give $\frac{1}{2}$ have $m = -\frac{1}{2}$ and $q^2 = 3/32$: the law's own $\sqrt{G\rho_\Lambda}$ times one of three rational halves (Misner–Sharp, Komar, $\sqrt{}$ of Bekenstein's $\frac{1}{4}$).
- **No evidential weight.** The class covers the data band with probability 1.0000, for both classes and both bands, and for the 11-value minimal subclass. No member fixes a second, coefficient-dependent number: it reads either H_Λ (constant a_0) or $H(z)$, giving $a_0(2.5)/a_0(0) = 3.7687$, the record's rival law. Nearest class values: 0.51824 (canonical) and 0.61421 (alternative) without square roots.
- **Record error.** The record says the data admit only $R = 4$ among natural caps. A19's own grid is $R \in \{1, 2, 3, 4, 8, 12, 16\}$. On the CFG493 band ($\kappa = 0.417 \pm 0.095$), 5 of 7 lie within 2σ, and $R = 8$ (-0.67σ) is closer than $R = 4$ ($+0.87\sigma$). Erratum e1fd7be4dc.

- **Scope.** Classes outside this one (bare-G portals, media, quantised couplings) stay open. The data cannot rank $\frac{1}{2}$ against $Z/4\pi$.

3 The strong-field turn-off and Cassini

3.1 The turn-off has no mechanism, and y_t is κ (D7)

- **The canonical law needs a turn-off.** Its tail is $y(\nu - 1) \to 1$, a constant anomaly a_0. At 1.5 AU that is 2557.4 times the planetary bound B_E on the canonical footing and 3087.4 times on the alternative. The Galaxy's external field cannot remove it: the monopole of the phantom source changes by less than 10^{-9}.
- **y_t is κ relabelled.** The record's closure $I(y_t) = 8\pi/\kappa^2$ is one-to-one: $y_t = 66.2999$ (quoted Λ), 66.3803 (32π), 46.0148 (alternative), 77.876 (thermal). The falsifier band $y_t \in [60, 140]$ is exactly $\kappa \in [0.3415, 0.5267]$. Nothing measures y_t now: at $y_t = 140$ Saturn's signal is 5.5×10^{-4} of its bound, and wide binaries move by at most 0.57%.
- **No mechanism in the action.** The stiffness term is a mass threshold, not an acceleration threshold, and curing the planets that way is the medium-first reading. No gate potential $W(u)$ can produce the turn-off: the anomaly peaks at $y = 12.24$ with $u = 0.899\,a_0$, and the same u recurs at $g = 0.816\,a_0$ and $72.9\,a_0$. None of 48 frozen scale monomials matches y_t on all three points; a disclosed coincidence, 66.46, is off by 44% on the alternative footing.
- **Record errors.** The cure's necessity was not open (A4 had it); the "planets pass 9.14×" margin rests on a stand-in Neptune bound; an $O(10^2)$ external-field suppression cannot act on a $1/y$ tail; the thermal y_t is 77.876, not 77.85 or 77.944. Erratum 3c11f5d2e0.

3.2 Cassini Q_2 of the unscreened canonical law (D7b, E2)

D7b computed the Solar-System quadrupole Q_2 of the canonical law in QUMOND, without stiffness screening (the DSF-21 A26 "law-first" reading), with and without the turn-off, for four external fields (A60's framework Milky Way, and kinematic totals 1.9, 2.2 and $2.6\,a_0$).

- **With the action's field equation (E2, AQUAL): the unscreened canonical law fails by 2.44–4.59×** the Hees line (canonical) and 2.85–5.19× (alternative), at every y_t row; AQUAL/QUMOND is 1.23–1.49 depending on the external field. These values supersede D7b's QUMOND values below. E2's AQUAL solver reproduces Blanchet & Novak's AQUAL benchmark to 1.1%.
- **D7b (QUMOND, superseded): the unscreened canonical law fails Cassini by 1.7–4×.** Against the Hees et al. (2014) 2σ line ($9.0 \times 10^{-27}\,\mathrm{s}^{-2}$) it fails at every y_t, every external field and both footings: 1.70–3.70× (canonical) and 1.91–4.03× (alternative). It also fails the tighter Park et al. (2026) line. The most favourable field still fails (1.70× canonical).
- **The turn-off cannot help.** It changes Q_2 by +0.06% to +0.69% (upward). Across the band $y_t \in [60, 140]$, Q_2 moves by at most $1.6 \times 10^{-28}\,\mathrm{s}^{-2}$, 0.052σ of Hees et al. So no κ range is selected. Q_2 is set near $y \approx 1$ ($r \approx 7000$ AU); the turn-off acts at $y > 60$.
- **The class conflict.** This failure is the known conflict between the RAR and the Solar-System quadrupole in this class of theories, already shown by Desmond, Hees & Famaey (2024). It is not new, and it is not a property of the turn-off. The statement is about the unscreened law only; the screened case is D7d.
- **Checks.** An exact limit derived for this lane, $q = -(16/49)\sqrt{e_N}$, is reproduced to 5×10^{-10}; two routes agree to 1.1×10^{-3}; our QUMOND value is 0.737 of Blanchet & Novak's AQUAL value (AQUAL fails harder). **K4 fails as frozen**: against Desmond et al.'s anchors our $|q|$ is 0.974, 0.965 and 0.965 of them (tolerance 3%); the gap is not a tail effect, and a lower q only strengthens the failure. The script therefore exits 1 (7/8). The record's "tail" explanation of a 1.03 calibration factor is wrong.

3.3 Cassini Q_2 of the screened theory (D7d)

D7d solved the stiffened action (A33 (L) and (P), static, $O(\varepsilon)$) for the Sun in a uniform Galactic field, with a linear Fourier route and a nonlinear grid route. Its $L_\star \to 0$ limit is AQUAL with $\mu = x/(1+x)$: it reproduces Blanchet & Novak's AQUAL value (3.818×10^{-26} against 3.8×10^{-26}). Criteria a13e418498, pre-result amendment e1ba64cf92, result b26f14bb82. CRISP FINDING; 10/10 checks.

- **The screened theory passes Cassini by 10^7–10^9.** At $L_\star = 24\,\mathrm{pc}$, $Q_2 = 1.0$–$2.3 \times 10^{-34}\,\mathrm{s}^{-2}$ over the four fields and both footings; at 90 pc, 1.9–4.3×10^{-36}. That is 2.2×10^{-10} to 2.5×10^{-8} of the Hees line. It passes Park et al. too.
- **Mechanism.** Inside $L_\star$ the Sun's **P** field is a constant hedgehog $GM/L_\star^2\,\hat{r}$, which has no quadrupole. Q_2 comes only from the anisotropic Galactic term acting at $r \sim L_\star$, so $Q_2 \propto GM/L_\star^3$. The stiffness is not a multiplicative screening factor in an external field.
- **The Cassini threshold.** The theory passes for $L_\star \geq L_{\star C}$ with $L_{\star C} \leq 0.069\,\mathrm{pc}$ (canonical) and $\leq 0.111\,\mathrm{pc}$ (alternative), at all four fields. This is an upper bound on the threshold, not a measurement: between 0.03 and 1 r_M the solver did not converge, and $L_\star = 0$ fails. The binding lower gate stays CFG691's Saturn monopole (5.75 pc).
- **Consequences (reported only).** At $L_\star = 24\,\mathrm{pc}$ the inner-planet anomaly is $GM/L_\star^2 = 2.42 \times 10^{-16}\,\mathrm{m\,s}^{-2}$ (Mars at 0.17 of its bound), and a 20 kAU wide binary has $\gamma_v - 1 = 8.1 \times 10^{-6}$. The unscreened law-first reading predicts $\gamma_v \approx 1.17$–1.20 and fails Cassini. The two readings cannot both hold, and Gaia DR4 separates them.
- **Limits.** Uniform, static Galactic field over $\sim L_\star$; $O(\varepsilon)$; no ephemeris refit. The main run peaked at 3.0 GB, over the 2 GB budget (disclosed); the mutation run covers four checks (2/4, it bites). Record notes: CFG770's negative sign is the Φ convention; CFG832's 0.087 pc gate was an extrapolation of the right order.

3.4 The Q_2–RAR gate: does any unscreened law pass both? (D7c)

D7c scored the record's whole unscreened law family against both Cassini Q_2 (D7b's solver, Hees 2σ, all four external fields) and SPARC (AG4b's per-galaxy pipeline, paired statistic, fail at $z \leq -2$). Criteria e9e5349368, result 26b5db080f. SCOPED NO-GO for the unscreened family, with one near-miss.

- **No law passes both on both footings.** The weakest tension anywhere in the family is 2.56σ (Hees), in one cell of the β family.
- **Named laws.** All five named record laws (the canonical $n = 1$ law, $n = 2$, ν_{mono}, the standard μ, ν_{RAR}) fail unscreened Q_2 at every external field on both footings: 1.18–4.15× the Hees line overall. All also fail Park et al.
- **The action's capacity family.** SPARC passes on both footings only at $n = 2$, whose Q_2 fails at 2.89–8.24σ. Q_2 passes at all four fields only at $n = 8$, which SPARC disfavours at $z = -6.32$ (canonical).
- **The near-miss.** In the β family (the record's "effective description"), $\beta \geq 2.5$ passes Q_2 on the canonical footing and the graded SPARC statistic, but fails the alternative footing in the framework's own Milky-Way field by 1.19–1.23× for β from 4 to 40, a floor no β crosses. The secondary SPARC statistic disfavours the same members; which statistic one trusts decides the canonical reading.
- **External-field dependence.** The canonical law passes the Hees line only for a total external field $g_e \leq 8.20 \times 10^{-11}$ (canonical) or $8.13 \times 10^{-11}\,\mathrm{m\,s}^{-2}$ (alternative), below both the kinematic field at the Sun and the framework's own Milky-Way field.
- **Under AQUAL (E2) the no-go strengthens.** No member passes Q_2 at all four fields on both footings. The Q_2-only passer $n = 8$ now fails (alternative E3 cell, 1.04×). The β members also fail the canonical E0 cell, by 1.01–1.12×: AQUAL raises their Q_2 by 1.03–1.22. The least-tense SPARC passer is $\beta = 5$, worst cell 1.41×.
- **Reconciliation with E1 (not papered over).** E1 redid D7c on the framework-native background (a_0 within 3% of canonical; no alternative footing) and found that the $\beta \geq 3$ members pass Hees at all four fields and SPARC's primary statistic at all three native a_0 (and $\beta = 2.5$ at two of three), at Q_2 of 0.87–0.99 of the line and SPARC z of -1.59 to -2.00: D7c becomes UNDECIDED, with the secondary SPARC statistic still against them ($z \leq -5.39$). But E1 used QUMOND. E2's AQUAL raises the same members' Q_2 by 1.03–1.22, more than E1's margins, and at canonical a_0 they fail E0 by 1.01–1.12×. Neither lane computed AQUAL at the native a_0. The combined status of the $\beta \geq 2.5$ members is therefore knife-edge on both sides: E1's Q_2 margins (0.87–0.99 of the line) are smaller than the AQUAL increase, and their SPARC passes sit within 0.41 of the $z = -2$ line. It is not a pass, and it is not a settled fail.
- **Checks.** 5/6: K4 fails as frozen (a tabulation difference of 3.3×10^{-9} against a 10^{-10} tolerance; numerical only), and the script exits 1. The first run's K3 failure was check-side round-off, fixed post-result (amendment A1); that run is kept. The mutation run bites (3/6). The statements concern unscreened law forms only; the screened case is D7d.

4 The stiffness length $L_\star$ (D8, D8b, E4)

- **Two objects, one length.** A76's sound speed (at the cosmic mean density) is exactly AG6D's free-medium equation of state (0.720398 m/s at 24 pc), so AG6D's bound applies: in the free medium $\sigma_c \leq 1.70$ (canonical) or 1.87 km/s (alternative), below the 10^4 K width by 4.84 and 4.41 times. $L_\star = \sigma_c^2/a_0$ as a free-medium stiffness is excluded; it survives only as a property of a tethered state that no action term yet defines.
- **Record error: the 24 pc floor is a measurement.** It is CFG820's globular-cluster χ^2 bound (fine-grid roots 23.641 / 24.476 pc). It excludes the thermal $L_\star$: $\Delta\chi^2 = 3.913$ (canonical, by 0.07) and 5.992 (alternative). Erratum 9e6fbbbfcd.
- **Measured dispersions.** Six tidal dwarfs give median $L_\star$ 42.2 / 35.0 pc with an 84/16 spread of 4.45. Stilp et al. (2013), 23 dwarfs at a common 200 pc beam, give median $\langle\sigma_{m2}\rangle = 9.30$ km/s, so $L_\star = 29.95$ / 24.80 pc, with 84/16 = 1.995, a knife-edge pass of the frozen "one constant" line (≤ 2). The narrow-peak width gives 20.5 / 17.0 pc, below the floor.
- **But σ is not universal (D8b).** Star formation does not drive it (FLOOR FAVOURED: $\rho = +0.474$, $p = 0.023$ against $\alpha = 0.0125$; slope +0.088, 95% CI [+0.026, +0.149], excluding 1/3). But its spread is real: $\chi^2 = 99.3$ for 22 degrees of freedom on tabulated errors, intrinsic scatter 0.96 km/s, against an 84/16 of 1.28 from noise alone. It tracks the H I column ($\rho = +0.613$, $p = 0.0024$). Read as a floor, the minimum σ_{m2} gives 17.0 / 14.1 pc, below the globular bound on both footings. So D8b does not strengthen the conditional prediction $L_\star \approx 25$–30 pc: it holds only if σ_c is a typical, gas-set value, and then $L_\star$ depends on environment, in conflict with the record's use of one $L_\star$ for gas-free systems.
- **The floor itself is soft (E4).** CFG820 built the 24 pc floor with the RAR fitting function and an additive external-field recipe, neither of which is the screened theory's physics. With the theory's own law and no recipe the floor is 28.85 pc (canonical) / 29.12 pc (alternative); with the law and the recipe kept, 23.87 / 24.66 pc, so the recipe, not the kernel, moves it. The action's own external-field response lies between and was not solved. The one stellar M/L shifted by its ± 0.15 dex spans 20.0–42.3 pc (canonical): the floor is calibration-dominated. Under the native model the thermal $L_\star$ is excluded at $\Delta\chi^2 = 6.21$ / 9.01. CFG821's dSph ceiling stays 22.0 pc, but it assumes no bound cold mass, which the screened theory requires for dwarfs; read literally the window [28.9, 22] pc would be closed.
- **Two verdicts above flip under clean inputs (E1).** D8's thermal exclusion ($\Delta\chi^2 = 3.91$, a 0.07 margin) does not survive a native a_0 (3.78, allowed) or a -0.2 dex M/L (0.32); D8b's floor reading flips both ways with the same M/L shift. Neither is a firm verdict.

5 No covariant parent (D9, D9b, E5)

- **U(1)-dielectric parents (D9).** Any such parent that enforces the law's Gauss identity (phantom = cold medium) has a fastest speed $v_{\max}^2 \geq (1 + g/a_0)/\Lambda_{\rm LV}$, where $\Lambda_{\rm LV}$ is the dark sector's Lorentz-violation factor. It is exactly 1 for every Lorentz-invariant $L(X, Y)$, so all such parents are superluminal (the record's NG2, now for every $L(X, Y)$; erratum 05a3bfdb73). A foliated parent is causal only if $\Lambda_{\rm LV} \geq 1 + g/a_0$: 6.3355×10^7 (canonical) and 5.2478×10^7 (alternative) at 1 AU. The frozen exact-speed checks U2, U4 and U5 failed (a parity-odd term mixes polarisations; 9/12); a post-result amendment proved the bound survives the mixing (3/3). A map of all parent classes is in the lane (`MAP.md`).
- **The record's action escapes only by not enforcing the identity.** DSF-21's A33 action is causal because its field is the capped u; its cone excess at $\eta = \gamma$ is 1.19×10^{-18} / 1.73×10^{-18} at $L_\star = 24$ pc. Its identity is a matching, not a field equation.
- **Enforcing it (D9b).** With a multiplier enforcing $\mathbf{D} \cdot \mathbf{P} = \rho_c - \bar{\rho}$, the literal action counts the medium twice ($s = 2$): at $g = a_0/3$ it misses the law by $-0.55\, a_0$, with no deep branch. The law-preserving version is unique, has 4 second-class constraints and 7 degrees of freedom, is causal and well posed on FRW, and is exactly AQUAL with $\mu = x/(1 + x)$ in the static limit. But it meets every hypothesis of the no-self-gravity theorem D-T (growth $a^{0.60953}$ in EdS), and the positivity no-go NG1 applies (ρ_c down to $-231\,\bar{\rho}$ at saddles, alternative). 14/14 checks; a first run with three code bugs (11/14) is kept.
- **With stiffness (E5).** The no-gos are a theorem for the stated classes, and they hold for the screened theory, not only at $L_\star \to 0$. At $L_\star \leq 90$ pc the stiffness moves the positivity no-go's saddle trough by at most 8.9×10^{-4} (cancelling it would need $L_\star \approx 3.2$ kpc for the Milky-Way–M31 pair); the no-self-gravity theorem is independent of the stiffness; a dark-sector curl stiffness only raises the superluminal speed or adds a ghost; and no stiffness or parameter-free gate removes the collapse-gate ghost. The superluminality is a positivity violation of the law's own Lagrangian (an external argument the lane cites, arXiv:hep-th/0602178). Open classes: a foliated U(1)

under the owner's criterion that a global time function suffices (D9's row rests on $v \leq c$); non-local parents beyond CFG947; non-variational latches; dispersion-held media; multi-field latches; parents that do not enforce the identity. 14/15: the frozen check E1 fails (its pass line omitted cross terms of a truncated series); a post-result amendment shows the residual is exactly those terms.
- **Conclusion.** Enforcing the identity is what brings back D-T and NG1, with or without stiffness. No covariant parent is established; $L_\star$ and η/γ remain inputs.

6 Tests and their power

6.1 The skew-sign test is retired (D1)

The record proposed the sign of the residual skew to separate $n = 1$ from $n = 2$. In $\log g$ both laws predict negative skew across the whole window ($\sqrt{N_s}$ skew at $a_0/2$: -4.38 for $n = 1$, -2.44 for $n = 2$). In linear g the signs differ, but errors symmetric in $\log g$ make a linear skew: $+0.42$ from a 0.06 dex background, and 2.86 (canonical) / 3.18 (alternative) measured on SPARC outside the window, against best-case signals of $+0.062$ and -0.164 (canonical), 17–51 times smaller. The record's power estimate (7,000–13,500 points) also used the $N_s = 1$ skew with $N_s = 93.1$ normalisation; correcting it multiplies the required number by 93 ($n = 1$) and 384 ($n = 2$). SCOPED NO-GO on any rotation-curve data. Erratum f775a7e6c5.

6.2 The wide-binary fork is decided by systematics (D2)

- **1.18 is the law's number.** An independent FFT QUMOND solve gives 1.1971 for A24's case (A24: 1.1933) and, orientation-averaged at the pre-registered external field, 1.175 (canonical) / 1.210 (alternative). Medium-first predicts 1.000.
- **With the action's field equation (E2).** AQUAL lowers γ_v by 0.007–0.011: 1.167 (canonical) / 1.199 (alternative) at the primary field. The registered Arm A band was built with QUMOND-type machinery, and the AQUAL values sit near its lower edge. A pre-data amendment A1 to the pre-DR4 note (ee621a0898) adopts the AQUAL values and tightens the scoring condition to a systematic below 0.048 (canonical) / 0.058 (alternative), not 0.054 / 0.064; it can only move outcomes to UNDECIDED. The frozen pre-registration is untouched.
- **The κ leverage was overstated.** With the shared systematic and the external-field degeneracy, $\sigma(\kappa) = 0.073$, so the joint error goes $0.121 \to 0.077$, not 0.0605 (floor 0.048 at infinite N).
- **One bin cannot decide.** The DR3 frozen-cut catalogue has 43 of 6,210 pairs (0.69%) in 18–22 kAU, not A24's 7.41%. The whole-catalogue $\hat{\gamma}$ separates the law's floor from 1.000 at 5.04–5.85σ (canonical) and 5.98–6.95σ (alternative), but 3σ is unreachable at any N if the systematic exceeds 0.054 (canonical) / 0.064 (alternative); the record's inter-group spread is 0.174. A pre-DR4 note fixes this rule before the data (commit 8658130bd0; the frozen pre-registration is untouched).
- **Satellites fail both readings.** A26's law-first verdict omitted the host's external field; with it, the 45 sub-threshold satellites sit +1.25 dex above the law (5 of 45 within 0.5 dex). Current data are open. Erratum c749c5007d.

6.3 The sealed ALMA tests are expected to be unscorable (D4)

The two governing plans (U4_27928, public 2026-11-06; GS4_24110, public 2026-11-28; dates from the record's archive metadata, not re-checked) are harness-complete on mock data. The record's forecast puts the gate statistic at S/N 0.18–0.58 (U4) and 0.04–0.15 (GS4) against a required 5, with no taper branch, so the expected outcome on both dates is "G1 FAIL, upper limit", with nothing scored. With perfect data, U4 reaches 4.34σ (framework vs rival, governing kernel, canonical) only for $V_c \approx 240$–260 km/s and only while the 0.2 dex stellar-mass systematic is kept out; propagated, the separation is 0.00. GS4 cannot separate $n = 1$ from $n = 2$ ($\leq 0.51\sigma$ on every branch). The governing plans score only the $\nu_{\rm mono}$ kernel, so the canonical law has no governing reading; a kernel addendum is the owner's choice. One disc excludes no model.

6.4 The native 1024^3 shear run died of memory (D3)

CFG960's pre-registered SC1x run (canonical footing) was killed four times by its own frozen swap rule, three times at the same force call (step 89 of 150, $a = 0.19$); the process footprint reached 65.2 GB on a 64 GiB machine, against a pre-registered estimate of 43–47 GB. SC2x (alternative) never started. No scored snapshot exists: **no verdict on either footing**. Status when this draft was written: the chain stopped on 2026-10-22 at 01:01 local and has not been restarted; the ledger row still reads "STARTED". A restart under the frozen files is expected to die at the same step; any fix changes a frozen file and needs a dated pre-result amendment, or a machine with $\geq$ 96–128 GB.

7 The retained-fraction mechanism (D6)

The record's cluster lensing needs the law's phantom to supply a fraction $u = 0.4716$ of the observed excess inside splashback (P4). D6 looked for a mechanism. Criteria 3082c43c79, pre-score amendment daf92d2d11, result b21dda1a2d. CRISP FINDING, but no mechanism works.

- **The record's cluster model gives zero.** CFG959 sets its unrelaxed fraction to 1 for clusters, so P4's sourcing share is $1 - f_P = 0$, which is 13.0σ below 0.4716 (read the other way, $u = 1$, it is 14.6σ above). Getting 0.47 would need cluster galaxies with $v/\sigma = 1.64$. The two papers' "u" are complements under one symbol.
- **Record error: a sign.** At fixed retained fraction, $d\ln u/d\ln M_v = -1/6$ (numerically -0.16680), not the record's $+1/6$.
- **Stream counting gives $\frac{1}{2}$, and the profile rejects it.** If only inflowing cold matter sources the phantom (a new posit, not in the action), the self-similar infall solution gives $u \approx 0.50$ inside splashback at every accretion index tried, and $u_{\rm MID} = 0.489$ / 0.492 / 0.495 on the 318-cluster stack, within 0.7σ of 0.4716. But the 8-bin profile fails: $\chi^2 = 39.0$ / 37.3 / 36.7 for 8 degrees of freedom (canonical), the same as the record's own fitted 3D step, which the record already rejects. Its two second predictions also fail. Counting only the never-crossed stream gives $u_{\rm MID} = 0.06$–0.14.
- **Galaxy lensing gives a different u.** Isolated KiDS-1000 lenses give $u_{\rm gal} \geq 0.660 \pm 0.028$ (canonical) and 0.729 ± 0.031 (alternative), 6.8σ / 8.4σ above 0.4716 (statistical only; a -0.25 dex stellar-mass systematic would erase it). So $u \approx 0.47$ is not universal, but this exclusion does not survive clean inputs: with stellar masses -0.2 dex lower, $u_{\rm gal} = 0.501$, $+1.5\sigma$ (E1). Cluster hydrostatic masses give 0.344 or 0.230 depending on the bias assumption; the two ends differ by 6.1σ, so this channel cannot yet test u independently of lensing.
- **Disclosures.** K3 fails (the solver's interior slope is 0.625 against the asymptotic 0.75), so the deep profile is not certified and the script exits 1 (6/7). K4's tolerance was loosened after the run (an interpolation offset of 6×10^{-6}), disclosed in the code. The mutation run bites (5/7). Record error also noted: the record's "3D step in r" should read "a feature of the projected ratio".

8 Hidden inputs and the fitted constants (E1, E4)

Wave E applied the owner's data rule: every verdict compares raw measurements with the framework's own predictions, and ΛCDM-processed or standard-MOND ingredients are reported separately with the verdict's sensitivity to them.

- **Input ledger (E1).** 43 inputs, each classed raw, framework-native, survey calibration, ΛCDM-processed or standard-MOND machinery. Criteria 581d52210b, result 77a8b2d4a6; 6/6 checks.
- **The footings.** A framework-native H_0 (55.0–57.5 km/s/Mpc from the record's own background fits) gives a_0 within -1.4% to $+3.0\%$ of the canonical footing: the canonical footing is clean to 3%. The alternative footing has no native counterpart; it is a pure ΛCDM import. New flag: the native H_0 band sits 15.5–18.0σ below the local distance ladder on disk (73.0 ± 1.0), which SPARC's Hubble-flow distances use, so no single H_0 makes the framework's background, ρ_Λ and the SPARC distances agree.
- **Verdicts that flip.** Four do not survive clean substitution: D7c (UNDECIDED on the native background, with the reconciliation in Section 3), D8's thermal-$L_\star$ exclusion, D8b's floor reading, and D6's "$u_{\rm gal}$ excludes 0.47". The others survive (D1, D2, D4, D5, D6 items 1–3, D7, D7b, D8b items 1–2).
- **κ (E4).** With a ΛCDM-free background (BAO + BBN) the gas routes give $a_0 = 7.986 \times 10^{-11}\,{\rm m\,s^{-2}}$ and $\kappa = 0.414 \pm 0.094$ (canonical), a shift of -0.03σ: κ is robust to the background. What the data fix is a_0; writing it as κ adds the claim $a_0 \propto \sqrt{\rho_\Lambda}$. Criteria 9cc8fcae21, pre-result amendment 9b3dfb1000, result c5a9768e6a; 6/6.

- u **(E1, E4).** u depends on the distance convention and on a_0: $d\ln u/d\ln a_0 = +0.55$. At the native h the 0.47 benchmark becomes 0.643. At the galaxy-measured a_0, $u_{\rm MID} = 0.433 \pm 0.030$ (post-result, reported only), and the error a_0 carries into u (0.059) is the largest term in its budget. So u is not an independent measured fraction.
- **Independence.** The three constants are independent in the action (Jacobian rank 3, singular values 1.05, 0.62, 0.35); no candidate relation among them is forced, so there is no second prediction to test.

9 Dwarf satellites against the screened theory (E3)

Criteria 5686635887, result c5e4196bb9; 6/6.

- **The excess is real.** The 45 sub-threshold satellites sit $+2.01 \pm 0.12$ dex above Newton with statistical binary corrections ($+1.64 \pm 0.12$ in the worst clean combination).
- **But $+2.15$ dex is the wrong yardstick.** At $L_\star = 24$ pc the screening still gives these systems a median boost of $10^{1.23}$ over Newton. The data sit $+0.89 \pm 0.13$ dex above the screened prediction (alternative $+0.85$), and $+1.52$ dex at $L_\star = 90$ pc. The record's "largely screened and nearly Newtonian" is wrong.
- **The capacity is undecidable.** At the census retained fraction, the median system must hold $q = 0.81$ of its whole catchment's cold energy inside its half-light radius; census-free, $q = 8.0$, a shortfall of $+0.90$ dex. The census is ΛCDM-processed and the record holds no binding mechanism. UNDECIDABLE on both footings.

10 Record corrections

Six errata were written fix-forward into the record (originals kept): f775a7e6c5 (D1: skew-sign discriminator retired), e1fd7be4dc (D5: the data do not select $R = 4$), 3c11f5d2e0 (D7: turn-off needed; planet margin; thermal y_t), 9e6fbbbfcd (D8: the 24 pc floor is CFG820's bound; free-medium $L_\star$ excluded), c749c5007d (D2: κ leverage, bin occupancy, fork not resolved), 05a3bfdb73 (D9: NG2 covers all Lorentz-invariant $L(X, Y)$). The DSF-21 documents now call $\kappa = Z/4\pi$ a postulate and credit Milgrom (d3e4110528).

11 What this establishes, and what it does not

- **The screened theory.** It is the one version that passes both Cassini (by 10^7–10^9) and the galactic law. Its open items: the dwarf gap (+0.89 dex, capacity undecidable), a calibration-dominated $L_\star$ floor, an a_0- and h-dependent u, and no covariant parent in the stated classes. Its decisive test is Gaia DR4: $\gamma_v - 1 \approx 8.1 \times 10^{-6}$ against the unscreened reading's ≈ 1.17.
- **Established.** κ is not derivable in the horizon-thermodynamic class, and the data do not select $\frac{1}{2}$ among natural caps. The unscreened canonical law fails Cassini Q_2 by 2.44–5.19× with the action's AQUAL field equation (1.7–4× in QUMOND), and the turn-off cannot change that; no member of the unscreened law family passes both Q_2 and SPARC on both footings. The turn-off has no mechanism in the action. Free-medium $L_\star$ is excluded. Gauss-enforcing parents are superluminal, Lorentz-violating, or lose self-gravity. The skew-sign test is retired. No tested mechanism gives the retained fraction $u \approx 0.47$, and it is not universal.
- **Open and data-gated.** The wide-binary fork (Gaia DR4, if the systematic is below 0.048 / 0.058 after the AQUAL amendment). The ALMA tests (expected unscorable). The native shear test (no verdict). The β members of the unscreened family (knife-edge between E1 and E2). The dwarf capacity (needs a native catchment rule or isolated dwarfs).
- **Not derived.** $\kappa = \frac{1}{2}$, $L_\star$, η/γ, the retained fraction. No covariant parent is established. Cold energy remains a medium, not a particle species. No result here is evidence for the framework over its rivals.

Disclosures. D7b's frozen check K4 fails (anchor convention; exit 1) and one post-result amendment explains it. D9's frozen checks U2, U4, U5 fail; amendment A1 (post-result, 7bdd5fee57) adds the lower-bound checks; the first run is kept. D9b's first run (11/14, code bugs) is kept. D8's amendment A1 (ce9340ccbc) and A2 (95a7ec3a41, the owner-approved fetch of Stilp et al.) are pre-result. D8b's K4 label parser was fixed after its first run (no criterion changed). D7c's K4 fails as frozen; its amendment A1 is post-result (check-side round-off) and its first run is kept. D7d's amendment A1 is pre-result; its main run used 3.0 GB (over budget) and its mutation run covers four checks. E1's first run misparsed its ledger header (fixed before the scored run). E3's mutation run passes K6, against the criteria's expectation (disclosed by the lane). E4's u at the galaxy a_0 is post-result. E5's frozen check E1 fails; its

amendment A1 is post-result. D6's amendment A1 is post-prototype and pre-score; its K3 fails and its K4 tolerance was loosened after the run. D3 is read-only forensics with no criteria file. D1's best case is an upper bound on the signal, not a forecast. D4's power numbers are noise-free ceilings. Not blind: the lanes read the record's numbers before freezing.

Repository lanes (criteria, scripts, outputs) under `doors_push_2026_10_22/`: D1–D9b as in Table 1; the pre-DR4 note under `prep_2026/gaia_dr4_prep/`. Companion notes: PAPER49 v1.2 (doi:10.5281/zenodo.23533323), PAPER55 (doi:10.5281/zenodo.23482936), PAPER56 (doi:10.5281/zenodo.23532875), PAPER57 (doi:10.5281/zenodo.23532885). Planetary bounds: A. Hees, W. M. Folkner, R. A. Jacobson, R. S. Park, PRD 89, 102002 (2014); R. S. Park et al., PRD 114, 024066 (2026). Class conflict: H. Desmond, A. Hees, B. Famaey, MNRAS 530, 1781 (2024). Cross-check: L. Blanchet, J. Novak, arXiv:1105.5815 (2011). Data: S. A. Cookson et al., MNRAS 547, stag342 (2026) (wide binaries); A. M. Stilp et al., ApJ 765, 136 (2013) (dwarf H I dispersions); F. Lelli et al., A&A 584, A113 (2015) (tidal dwarfs); F. Lelli, S. McGaugh, J. Schombert, AJ 152, 157 (2016) (SPARC). Theory: M. Milgrom, ApJ 270, 365 (1983); M. Milgrom, Phys. Lett. A 253, 273 (1999) (astro-ph/9805346); M. Milgrom, arXiv:1605.07458; M. Milgrom, MNRAS 403, 886 (2010) (QUMOND). AI assistance: Anthropic's Claude (lanes, review, drafting).